VyapaarSetu — Solution Architecture

AI-POWERED MSE AGENT MAPPING TOOL FOR ONDC TEAM INITIATIVE

INPUT CHANNELS

Voice Call

LIVE

Web Browser Interface

Web Speech API
Sarvam AI STT / TTS

↓

WhatsApp

ASYNC

Baileys Service

WhatsApp Web Protocol
Port 3001

↓

Web Chat

INTERACTIVE

Next.js Frontend

React UI
Real-time Chat

ALL CHANNELS CONNECT TO

FastAPI Backend
Python · Port 8000

DUAL AGENT ARCHITECTURE

Agent Engine

FastAPI Backend · Orchestrator

JWT Authentication · RBAC (MSE · SNP · Admin)

ORCHESTRATOR ROUTES TO AGENTS

Voice Agent < 2s

- Single LLM call, no tools
- Fast response under 2 seconds
- MSE profile injected
- Mobile verification
- Field extraction via [FIELDS] tags

ReAct Agent REASONING

- Multi-turn reasoning loop
- Max 5 iterations
- Tool execution engine
- Skill discovery from prompts
- WhatsApp + Web channels

ReAct Agent connects to Skills + Tools

SKILL-BASED FRAMEWORK

EACH WITH SKILL.MD PROMPT

Onboarding	Step 1
Cataloguing	Step 2
Matching	Step 3

8 AGENT TOOLS

CALLABLE BY REACT AGENT

- classify_product
- extract_attributes
- validate_catalogue
- get_category_info
- match_snp
- fetch_udyam_data
- search_knowledge RAG
- send_notification

EXTERNAL SERVICES

DeepSeek LLM API

Model-agnostic
Swappable provider
Primary inference engine

Sarvam AI APIs

STT — Speech to Text
TTS — Text to Speech
LID — Language Detection

API calls from Agent Engine

Udyam API

MSE registration lookup
Business profile auto-fetch
(Mocked for PoC)

Voice Pipeline & RAG Pipeline

VOICE PIPELINE

User Speech
Microphone
VAD attached

→

Sarvam AI STT
Speech → Text
Hindi/English

→

Text Input
Transcribed
query

→

Agent Engine
Voice / ReAct

→

Response Text
Agent output
+ [FIELDS] tags

→

Sarvam AI LID
Language
Detect

→

Sarvam AI TTS
Text → Speech
Hindi/English

→

Audio Playback
User hears
response

RAG PIPELINE — ONDC KNOWLEDGE RETRIEVAL

Source Documents
ONDC Taxonomy JSON
TEAM Scheme Guidelines
SNP Knowledge Base

→

Text Chunking
Recursive splitting
Overlap strategy

→

Embedding
Sentence-Transformers
Local model

→

Qdrant Vector DB
Vector storage
+ BM25 index

search_knowledge tool Hybrid Search + Re-ranking Relevant ONDC Context ReAct Agent

AI CATALOGUING PIPELINE (CAAS LAYER)

Product Input
Voice / WhatsApp
/ Web text

→

classify_product
LLM maps to
ONDC RET code

→

extract_attributes
NER extracts
mandatory fields

→

validate_catalogue
Compliance check
E-Commerce Rules

→

ONDC Ready
100% compliance
Status: READY

Data Layer & MSE Onboarding Flow

DATA LAYER

PostgreSQL
Users · MSEs · SNPs
Products · OndcCategories
MSEMatches · Sessions · Messages
WhatsApp Auth State

Redis
Session Cache
Fast Reads
Rate Limiting

Qdrant
ONDC Taxonomy Vectors
SNP Knowledge Base
TEAM Scheme RAG
Hybrid Search Index

3-STEP MSE ONBOARDING PROCESS

STEP 1 — REGISTRATION & PROFILE

Login
Udyam + Mobile OTP

→

Udyam Fetch
Auto-populate business details

→

Voice Profile
Voice-enabled form filling

→

Profile Complete
MSE registered

STEP 2 — AI CATALOGUE BUILDING

Voice Call
Collect 6 fields per product

→

AI Pipeline
Classify + Extract + Validate

→

WhatsApp Follow-up
Collect missing fields async

→

Catalogue Ready
100% compliance

STEP 3 — SNP MATCHING & ONBOARDING

AI Matching
Category + Geography + B2B/B2C scoring

→

Top 3 SNPs
Ranked with match reasons

→

MSE Selects
Profile + catalogue shared with SNP

→

Live on ONDC
SNP onboards MSE

3 PORTALS

MSE Portal

Registration · Voice Catalogue
SNP Matching · Dashboard
Knowledge Base (RAG)

SNP Portal

AI-matched MSE leads
Readiness scores
Accept/Reject workflow

NSIC Admin Portal

Funnel analytics dashboard
AI pre-screened verification
WhatsApp bot management

Voice → WhatsApp Handoff & Architecture Summary

VOICE → WHATSAPP HANDOFF FLOW

Voice Call
Collect product fields (6 per item)

→

POST /products/
API endpoint + voice_fields

→

AI Pipeline
Classify + Extract + Validate

→

WhatsApp Notification
Missing fields sent to MSE

→

MSE Replies
On WhatsApp one field at a time

→

ReAct Agent
Collects remaining fields via tools

→

Product Update
Database update

SNP MATCHING ALGORITHM — WEIGHTED SCORING

Category Overlap
45%
MSE categories vs SNP supported categories

+

Geography
25%
Target states vs SNP coverage

+

Transaction Type
20%
B2B/B2C alignment with SNP capabilities

+

Capacity
10%
Onboarding speed and capacity score

=

Match Score
Top 3
Ranked SNPs with reasons displayed

ARCHITECTURE SUMMARY

Input Channels	3 channels: Voice (Web Speech API + Sarvam AI), WhatsApp (Baileys, port 3001), Web Chat (Next.js)
Backend	FastAPI (Python, port 8000) with JWT Auth + RBAC for MSE, SNP, Admin roles
Agent Engine	Dual architecture: Voice Agent (single LLM, <2s) + ReAct Agent (multi-turn, 5 iter, tool use)
Skills	3 skill prompts: Onboarding, Cataloguing, Matching — each with dedicated tools
Tools	8 tools: classify_product, extract_attributes, validate_catalogue, get_category_info, match_snp, fetch_udyam_data, search_knowledge (RAG), send_notification
RAG Pipeline	ONDC taxonomy + TEAM scheme + SNP data → Chunking → Sentence-Transformers → Qdrant → Hybrid Search + Re-ranking
Voice Pipeline	Speech → Sarvam STT → Agent → Sarvam LID → Sarvam TTS → Audio (with VAD on mic input)
Data Layer	PostgreSQL (MSEs, SNPs, products, matches), Redis (cache), Qdrant (vectors)
External APIs	DeepSeek LLM (model-agnostic, swappable) + Sarvam AI (STT, TTS, LID) + Udyam API